📊 Slide 1 — Data Acquisition & Quality

Source, Period, Quality Issues & Resolutions

DATA SOURCE & PERIOD

- Provider: Yahoo Finance (yfinance Python library)
- Period: 01 Jan 2023 – 31 Dec 2024 (2 full calendar years)
- Frequency: Daily OHLCV (Adjusted Close for all return calculations)
- Universe: 15 NSE-listed stocks across 3 sectors

  Banking (5): HDFCBANK, ICICIBANK, SBIN, KOTAKBANK, AXISBANK

  IT (5): TCS, INFY, WIPRO, HCLTECH, TECHM

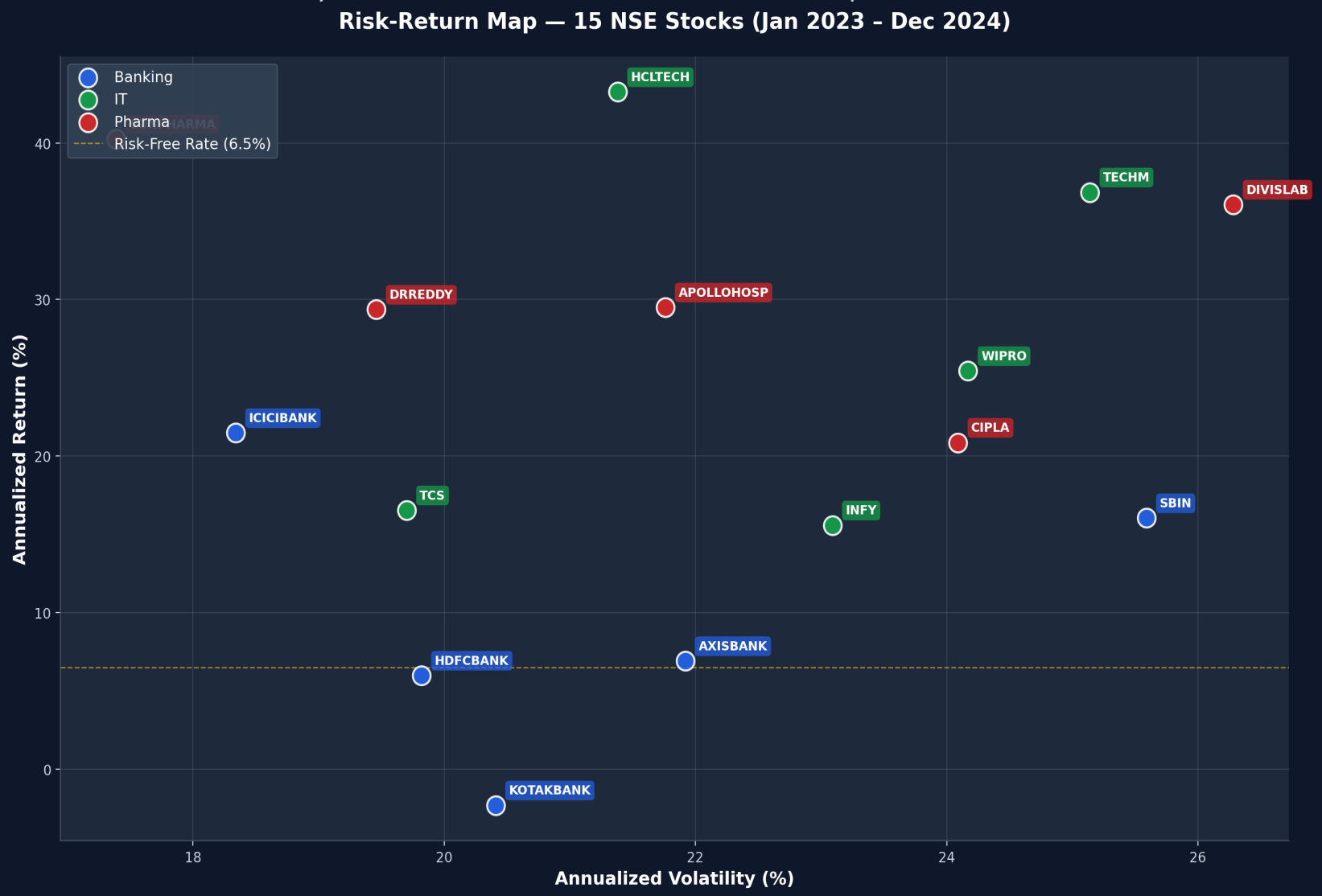
  Pharma (5): SUNPHARMA, DRREDDY, CIPLA, DIVISLAB, APOLLOHOSP
- Benchmark: Nifty 50 Index (^NSEI)

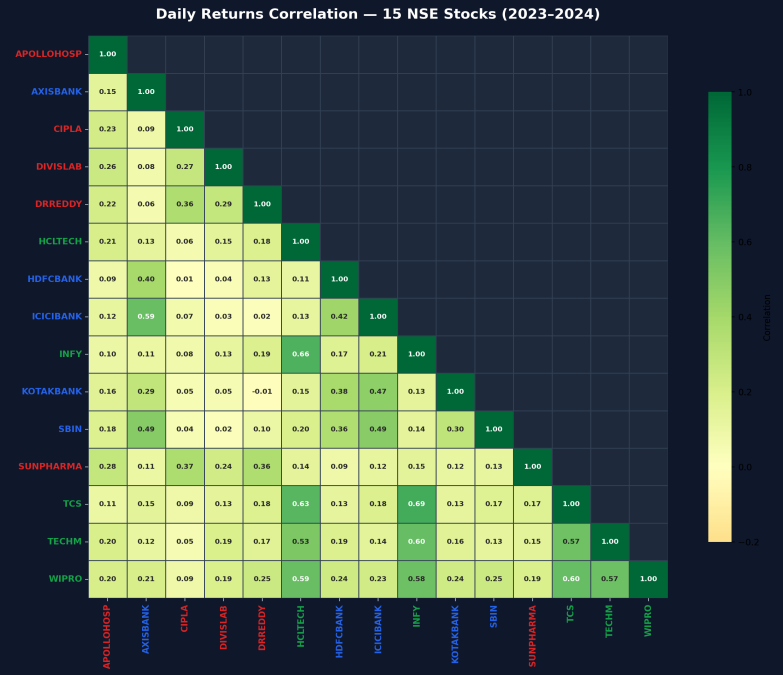
DATA QUALITY ACTIONS

- Missing values → Forward-filled, then back-filled
- Indian market holidays → Expected gaps, no additional treatment
- Zero/negative prices → Replaced with forward-fill
- Extreme returns (>25% daily) → Flagged but retained (legitimate events)
- Validated: No stock has >5 consecutive missing trading days
- All timestamps timezone-normalized (IST assumed)

📈 Slide 2 — Risk-Return Map

Annualized Volatility vs Annualized Return — Sector Clusters Explained





Daily Returns Correlation Heatmap

KEY OBSERVATIONS

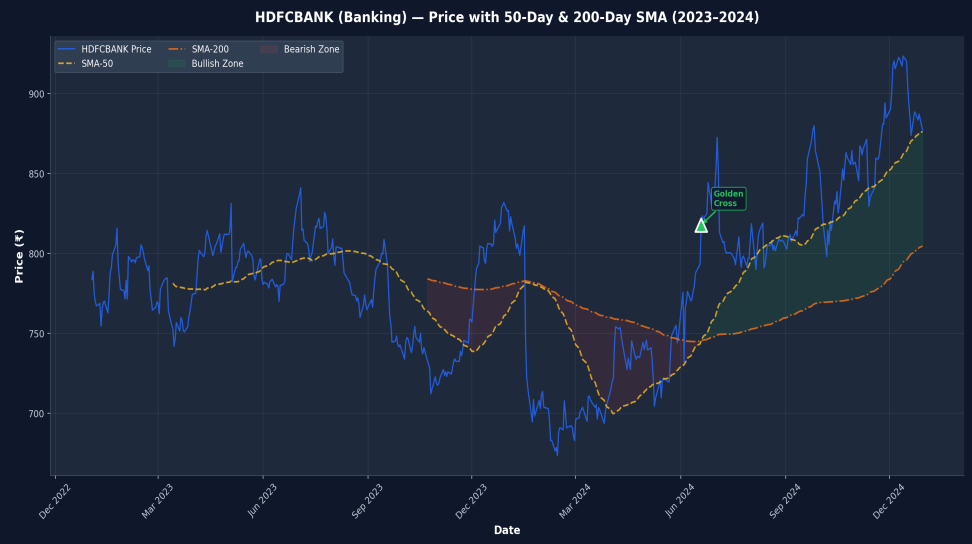
- Banking cluster: Moderate vol, correlated
- IT cluster: Varies widely in risk-return
- Pharma: Defensive, lower Beta
- Risk-free rate line at 6.5% p.a.

🏆 Slide 3 — Top & Bottom Stocks by Sharpe Ratio

Best 3 and Worst 3 performers — What separates winners from laggards

📉 Slide 4 — SMA Technical Signals

50-Day & 200-Day SMA — Golden Cross / Death Cross as of 31 Dec 2024

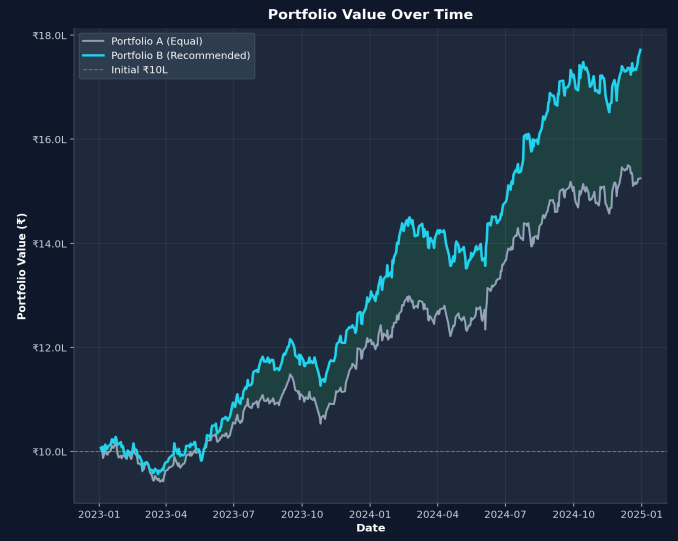


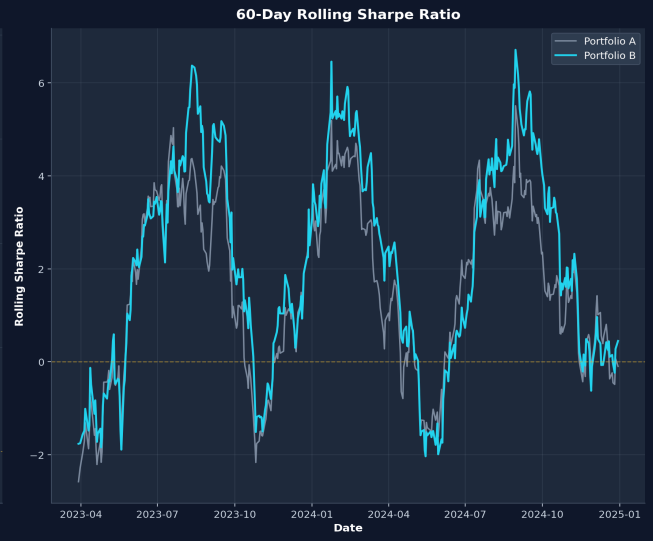
HDFCBANK — SMA Overlay (Banking)

TCS — SMA Overlay (IT)

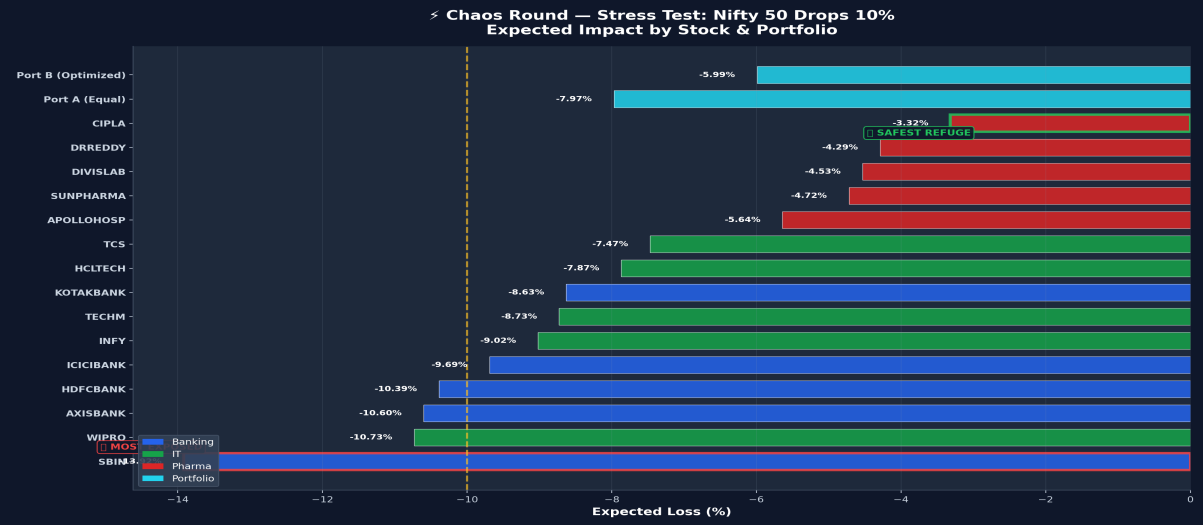
💼 Slide 5 — Portfolio Recommendation + Chaos Round

Portfolio A (Equal) vs Portfolio B (Optimized) + Nifty -10% Stress Test





⚡ Chaos Round: Nifty -10% Stress Test



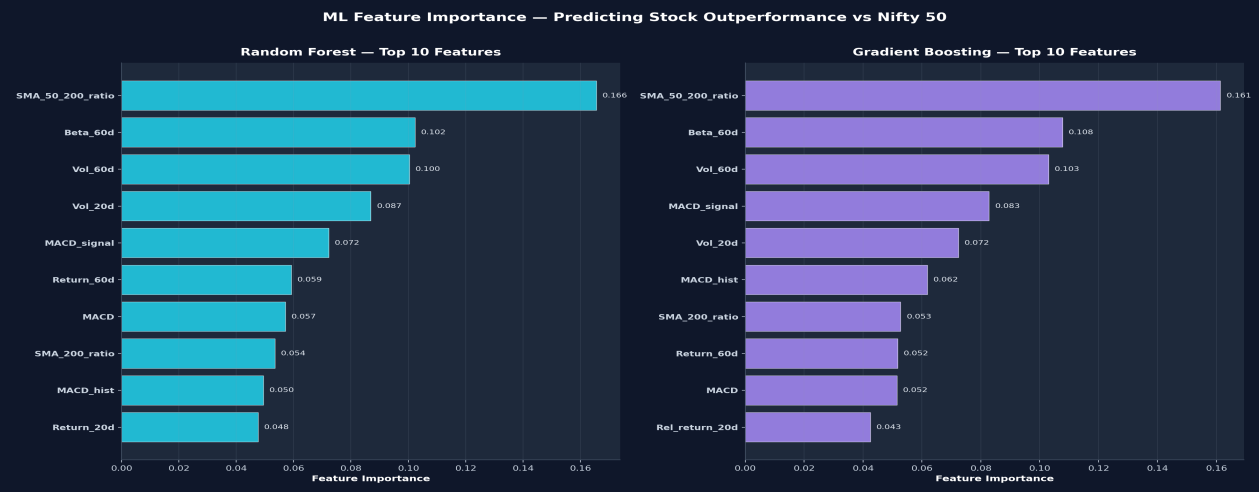
Portfolio B is strategically allocated to minimize drawdown using a defensive pivot.

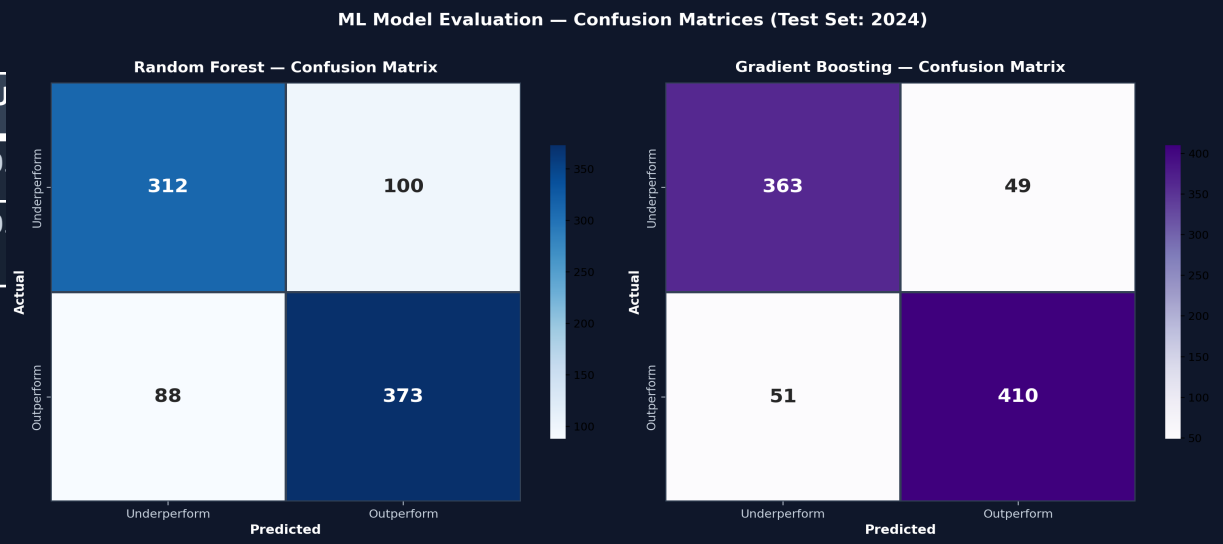
🤖 Slide 6 — ML Model Comparison

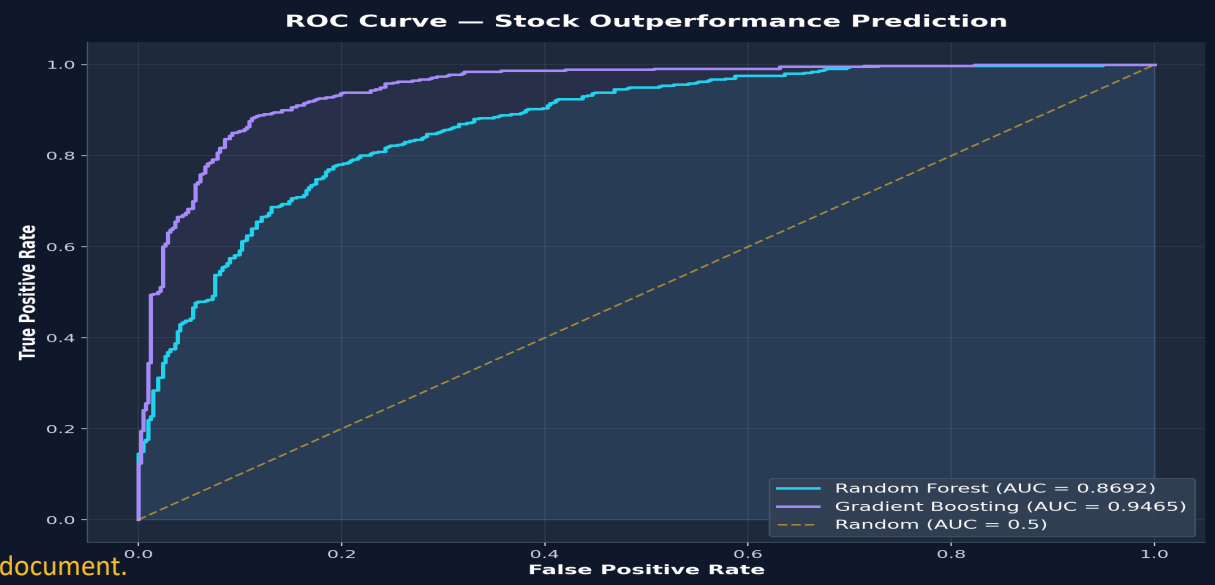
Random Forest vs Gradient Boosting — Predicting Stock Outperformance vs Nifty 50 (20-day forward)

Model Performance (Test: 2024)

Model	Accuracy	Precision	Recall	F1 Score	AU
Random Forest	0.7847	0.7886	0.8091	0.7987	0.
Gradient Boosting	0.8855	0.8932	0.8894	0.8913	0.







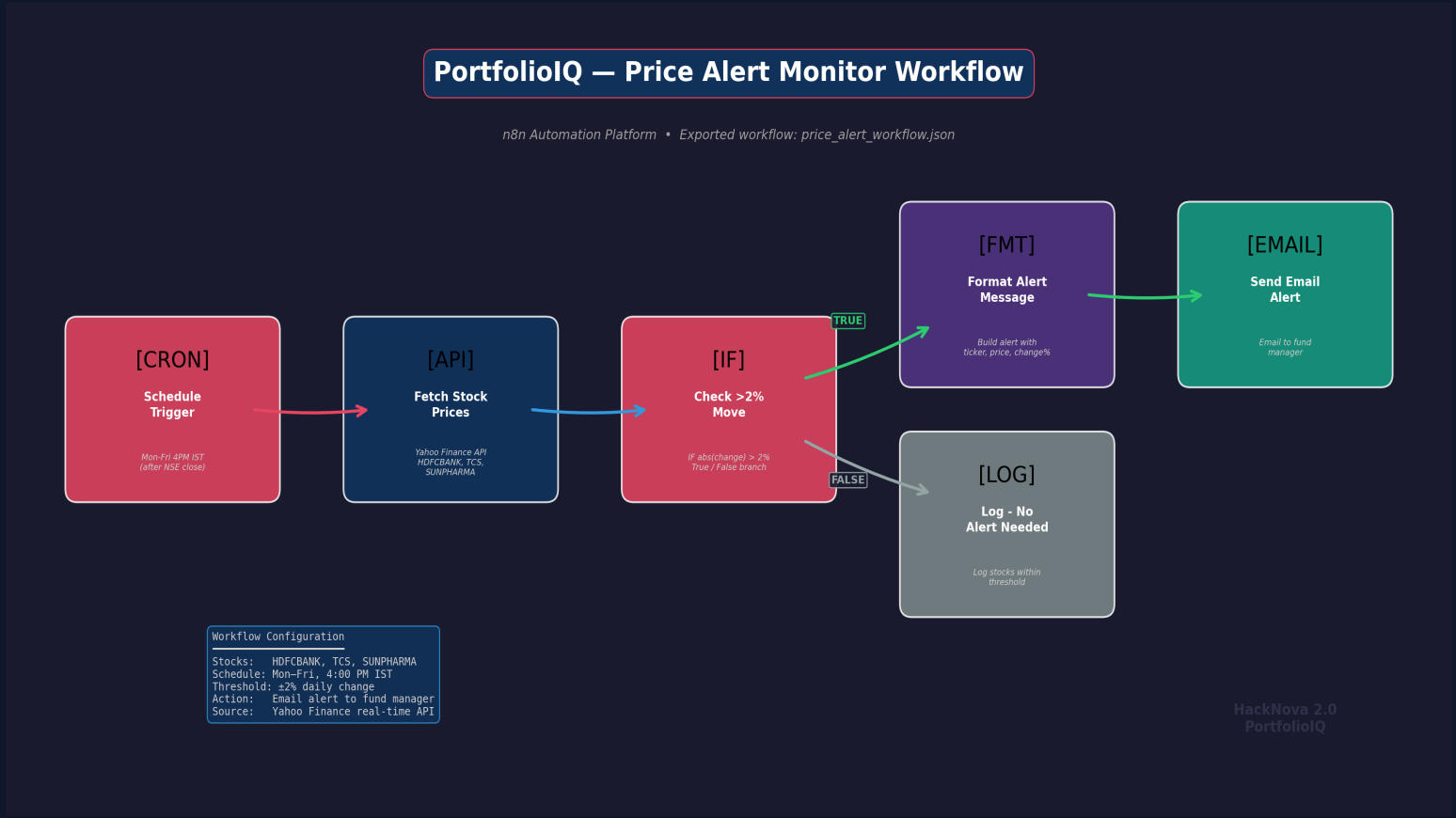
ML Design:

- Target: Does stock outperform Nifty 50 over next 20 days?
- Features: 18 technical indicators (returns, volatility, SMA ratios, RSI, MACD, Beta, volume)
- Temporal split: Train on 2023, Test on 2024 (no data leakage)
- Class balancing: balanced weights (RF) / scale_pos_weight (XGB)

Top 3 Features: Rel_return_20d, Vol_60d, MACD_hist. Financial interpretation provided in separate document.

⚙️ Slide 7 — n8n Automation Demo

Price Alert Workflow — Live Execution



WORKFLOW: PRICE ALERT MONITOR

Purpose:

Automated monitoring replaces manual daily checks, ensuring the fund manager is immediately notified of significant price movements requiring action.

Configuration:

- Stocks: HDFCBANK, TCS, SUNPHARMA
- Schedule: Mon–Fri at 4:00 PM IST (post NSE close)
- Threshold: Alert if any stock moves >2%
- Action: Email alert with ticker, price, change %

📺 **LIVE DEMO:**

Import JSON → Execute workflow → Show alert output

Workflow JSON → n8n/price_alert_workflow.json | Runs Mon-Fri 4 PM IST | Uses Yahoo Finance API | Sends alert via email to fund manager